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1 多场景英语教学音频播放与管控软件 V1.0
2 [padding]
3 # apps/api/main.py
4 from __future__ import annotations
5 [padding]
6 import os
7 import sys
8 from pathlib import Path
9 [padding]
10 import uvicorn
11 from fastapi import FastAPI
12 from fastapi.middleware.cors import CORSMiddleware
13 from fastapi.staticfiles import StaticFiles
14 [padding]
15 # When started via `python apps/api/main.py`, ensure repo root is
16 importable.
17 if __package__ is None or __package__ == "":
18     repo_root = Path(__file__).resolve().parents[2]
19     sys.path.insert(0, str(repo_root))
20 [padding]
21 from apps.api.routers import auth, control, devices, lessons, scenes, stats
22 [padding]
23 [padding]
24 def create_app() -> FastAPI:
25     app = FastAPI(title="多场景英语教学音频播放与管控软件 API", version=
26 "1.0")
27     app.add_middleware(
28         CORSMiddleware,
29         allow_origins=["*"],
30         allow_credentials=True,
31         allow_methods=["*"],
32         allow_headers=["*"],
33     )
34 [padding]
35     app.include_router(auth.router, prefix="/api")
36     app.include_router(scenes.router, prefix="/api")
37     app.include_router(devices.router, prefix="/api")
38     app.include_router(lessons.router, prefix="/api")
39     app.include_router(control.router, prefix="/api")
40     app.include_router(stats.router, prefix="/api")
41 [padding]
42     @app.get("/api/health")
43     def health() -> dict[str, str]:
44         return {"status": "ok"}
45 [padding]
46     app.mount("/", StaticFiles(directory="apps/web", html=True), name="web")
47     return app
48 [padding]
49 [padding]
50 app = create_app()
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51 [padding]
52 [padding]
53 def main() -> None:
54     port = int(os.environ.get("APP_PORT", "8000"))
55     uvicorn.run("apps.api.main:app", host="127.0.0.1", port=port, log_level=
56 "warning")
57 [padding]
58 [padding]
59 if __name__ == "__main__":
60     main()
61 # apps/api/routers/_init_.py
62 [padding]
63 # apps/api/routers/auth.py
64 from __future__ import annotations
65 [padding]
66 from fastapi import APIRouter, HTTPException
67 [padding]
68 from packages.core.auth import authenticate_user
69 [padding]
70 router = APIRouter(tags=["auth"])
71 [padding]
72 [padding]
73 @router.post("/login")
74 def login(payload: dict) -> dict:
75     username = str(payload.get("username", "")).strip()
76     password = str(payload.get("password", "")).strip()
77     user = authenticate_user(username=username, password=password)
78     if not user:
79         raise HTTPException(status_code=401, detail="invalid_credentials")
80     return {"token": f'demo-token: {user["id"]}', "user": user}
81 # apps/api/routers/control.py
82 from __future__ import annotations
83 [padding]
84 from fastapi import APIRouter, HTTPException
85 [padding]
86 from packages.core.store import db
87 [padding]
88 router = APIRouter(tags=["control"])
89 [padding]
90 [padding]
91 @router.post("/lessons/{lesson_id}/start")
92 def start_lesson(lesson_id: str, payload: dict | None = None) -> dict:
93     session = db.start_session(lesson_id=lesson_id)
94     if not session:
95         raise HTTPException(status_code=404, detail="lesson_not_found")
96     return {"session": session}
97 [padding]
98 [padding]
99 @router.post("/sessions/{session_id}/end")
100 def end_session(session_id: str, payload: dict | None = None) -> dict:
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101     session = db.end_session(session_id=session_id)
102     if not session:
103         raise HTTPException(status_code=404, detail="session_not_found")
104     return {"session": session}
105 [padding]
106 [padding]
107 @router.post("/sessions/{session_id}/event")
108 def post_event(session_id: str, payload: dict) -> dict:
109     action = str(payload.get("action", "")).strip()
110     actor_user_id = str(payload.get("actor_user_id", "")).strip()
111     target = str(payload.get("target", "all_students")).strip()
112     if not action or not actor_user_id:
113         raise HTTPException(status_code=400, detail="missing_fields")
114     event = db.add_event(session_id=session_id, actor_user_id=actor_user_id,
115 target=target, action=action, payload=payload.get("payload", {}))
116     if not event:
117         raise HTTPException(status_code=404, detail="session_not_found")
118     return {"event": event}
119 [padding]
120 [padding]
121 @router.get("/sessions/{session_id}/events")
122 def list_events(session_id: str, since: int = 0) -> dict:
123     items = db.list_events(session_id=session_id, since=since)
124     return {"items": items}
125 [padding]
126 [padding]
127 @router.get("/sessions/{session_id}/policy")
128 def get_policy(session_id: str) -> dict:
129     policy = db.get_session_policy(session_id)
130     if not policy:
131         raise HTTPException(status_code=404, detail="session_not_found")
132     return {"policy": policy}
133 # apps/api/routers/devices.py
134 from __future__ import annotations
135 [padding]
136 from fastapi import APIRouter
137 [padding]
138 from packages.core.store import db
139 [padding]
140 router = APIRouter(tags=["devices"])
141 [padding]
142 [padding]
143 @router.get("/devices")
144 def list_devices(scene_id: str | None = None) -> dict:
145     items = db.list_devices(scene_id=scene_id)
146     return {"items": items}
147 # apps/api/routers/lessons.py
148 from __future__ import annotations
149 [padding]
150 from fastapi import APIRouter, HTTPException
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151 [padding]
152 from packages.core.store import db
153 [padding]
154 router = APIRouter(tags=["lessons"])
155 [padding]
156 [padding]
157 @router.get("/lessons")
158 def list_lessons() -> dict:
159     return {"items": db.list_lessons()}
160 [padding]
161 [padding]
162 @router.post("/lessons")
163 def create_lesson(payload: dict) -> dict:
164     title = str(payload.get("title", "")).strip()
165     scene_id = str(payload.get("scene_id", "")).strip()
166     teacher_id = str(payload.get("teacher_id", "")).strip()
167     if not title or not scene_id or not teacher_id:
168         raise HTTPException(status_code=400, detail="missing_fields")
169     item = db.create_lesson(title=title, scene_id=scene_id, teacher_id=
170 teacher_id, book_unit=str(payload.get("book_unit", "")).strip())
171     return {"item": item}
172 [padding]
173 [padding]
174 @router.get("/lessons/{lesson_id}")
175 def get_lesson(lesson_id: str) -> dict:
176     item = db.get_lesson(lesson_id)
177     if not item:
178         raise HTTPException(status_code=404, detail="not_found")
179     return {"item": item}
180 [padding]
181 [padding]
182 @router.get("/lessons/{lesson_id}/clips")
183 def list_clips(lesson_id: str) -> dict:
184     return {"items": db.list_clips(lesson_id)}
185 [padding]
186 [padding]
187 @router.post("/lessons/{lesson_id}/clips")
188 def create_clip(lesson_id: str, payload: dict) -> dict:
189     label = str(payload.get("label", "")).strip()
190     start_ms = int(payload.get("start_ms", 0))
191     end_ms = int(payload.get("end_ms", 0))
192     asset_id = str(payload.get("audio_asset_id", "")).strip()
193     if not label or not asset_id or end_ms <= start_ms:
194         raise HTTPException(status_code=400, detail="invalid_clip")
195     item = db.create_clip(
196         lesson_id=lesson_id,
197         audio_asset_id=asset_id,
198         label=label,
199         start_ms=start_ms,
200         end_ms=end_ms,
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201         default_speed=float(payload.get("default_speed", 1.0)),
202         loop_suggest=int(payload.get("loop_suggest", 1)),
203     )
204     return {"item": item}
205 [padding]
206 [padding]
207 @router.post("/lessons/{lesson_id}/end")
208 def end_lesson(lesson_id: str) -> dict:
209     session = db.end_active_session(lesson_id=lesson_id)
210     if not session:
211         raise HTTPException(status_code=404, detail="not_found")
212     return {"session": session}
213 # apps/api/routers/scenes.py
214 from __future__ import annotations
215 [padding]
216 from fastapi import APIRouter, HTTPException
217 [padding]
218 from packages.core.store import db
219 [padding]
220 router = APIRouter(tags=["scenes"])
221 [padding]
222 [padding]
223 @router.get("/scenes")
224 def list_scenes() -> dict:
225     return {"items": db.list_scenes()}
226 [padding]
227 [padding]
228 @router.patch("/scenes/{scene_id}")
229 def patch_scene(scene_id: str, payload: dict) -> dict:
230     caps = payload.get("capabilities")
231     if not isinstance(caps, dict):
232         raise HTTPException(status_code=400, detail="invalid_capabilities")
233     item = db.update_scene_capabilities(scene_id=scene_id, capabilities=caps)
234 [padding]
235     if not item:
236         raise HTTPException(status_code=404, detail="not_found")
237     return {"item": item}
238 # apps/api/routers/stats.py
239 from __future__ import annotations
240 [padding]
241 from fastapi import APIRouter, HTTPException
242 [padding]
243 from packages.core.store import db
244 [padding]
245 router = APIRouter(tags=["stats"])
246 [padding]
247 [padding]
248 @router.get("/lessons/{lesson_id}/stats")
249 def lesson_stats(lesson_id: str) -> dict:
250     stats = db.lesson_stats(lesson_id=lesson_id)
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251     if not stats:
252         raise HTTPException(status_code=404, detail="not_found")
253     return {"stats": stats}
254 # packages/core/__init__.py
255 [padding]
256 # packages/core/auth.py
257 from __future__ import annotations
258 [padding]
259 from .crypto import hash_password
260 from .store import db
261 [padding]
262 [padding]
263 def authenticate_user(username: str, password: str) -> dict | None:
264     user = db.get_user_by_username(username)
265     if not user:
266         return None
267     if user["password_hash"] != hash_password(password):
268         return None
269     return {k: v for k, v in user.items() if k != "password_hash"}
270 # packages/core/crypto.py
271 from __future__ import annotations
272 [padding]
273 import hashlib
274 [padding]
275 [padding]
276 def hash_password(pw: str) -> str:
277     return hashlib.sha256(pw.encode("utf-8")).hexdigest()
278 [padding]
279 # packages/core/store.py
280 from __future__ import annotations
281 [padding]
282 import json
283 import time
284 import uuid
285 from dataclasses import dataclass
286 from pathlib import Path
287 [padding]
288 [padding]
289 def _now_ms() -> int:
290     return int(time.time() * 1000)
291 [padding]
292 [padding]
293 def _id(prefix: str) -> str:
294     return f"{prefix}_{uuid.uuid4().hex[:10]}"
295 [padding]
296 [padding]
297 @dataclass
298 class InMemoryDB:
299     base_dir: Path
300 [padding]
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301     def __post_init__(self) -> None:
302         self._users = []
303         self._scenes = []
304         self._devices = []
305         self._lessons = []
306         self._clips = []
307         self._sessions = []
308         self._events = []
309         self._seq = 0
310         self._seed()
311     [padding]
312     def _seed(self) -> None:
313         from .crypto import hash_password
314     [padding]
315         self._users = [
316             {"id": "u_teacher", "name": "英语教师", "role": "teacher",
317 "username": "teacher", "password_hash": hash_password("demo")},
318             {"id": "u_student", "name": "学生端", "role": "student",
319 "username": "student", "password_hash": hash_password("demo")},
320             {"id": "u_admin", "name": "管理员", "role": "admin", "username":
321 "admin", "password_hash": hash_password("demo")},
322         ]
323     [padding]
324         self._scenes = [
325             {"id": "scene_classroom", "name": "普通教室", "capabilities":
326 {"force_sync": False, "student_seek": False, "student_speed": False,
327 "student_loop": False}},
328             {"id": "scene_language_lab", "name": "语音室", "capabilities":
329 {"force_sync": True, "student_seek": False, "student_speed": False,
330 "student_loop": True}},
331             {"id": "scene_live", "name": "线上直播", "capabilities":
332 {"force_sync": True, "student_seek": False, "student_speed": False,
333 "student_loop": False}},
334             {"id": "scene_corner", "name": "口语角", "capabilities":
335 {"force_sync": False, "student_seek": True, "student_speed": True,
336 "student_loop": True}},
337             {"id": "scene_selfstudy", "name": "自习室", "capabilities":
338 {"force_sync": False, "student_seek": True, "student_speed": True,
339 "student_loop": True}},
340         ]
341     [padding]
342         self._devices = []
343         for scene in self._scenes:
344             sid = scene["id"]
345             self._devices.append({"id": "_id("dev")", "scene_id": sid, "name":
346 f'{scene["name']}' 教师控制台", "type": "teacher_console", "online": True,
347 "last_seen": _now_ms()})
348             for i in range(1, 4):
349                 self._devices.append({"id": "_id("dev")", "scene_id": sid,
350 "name": f'{scene["name']}' 学生终端 {i}', "type": "student_terminal",

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351 "online": (i != 3), "last_seen": _now_ms() - i * 7000})
352 [padding]
353     demo = self.create_lesson(title="Unit 1 听力热身", scene_id=
354 "scene_language_lab", teacher_id="u_teacher", book_unit="Unit 1")
355     asset_id = "asset_demo_001"
356     self._clips.extend(
357         [
358             {"id": _id("clip"), "lesson_id": demo["id"],
359 "audio_asset_id": asset_id, "label": "对话开头", "start_ms": 0, "end_ms":
360 12000, "default_speed": 1.0, "loop_suggest": 1},
361             {"id": _id("clip"), "lesson_id": demo["id"],
362 "audio_asset_id": asset_id, "label": "重点句复听", "start_ms": 12000,
363 "end_ms": 24000, "default_speed": 0.9, "loop_suggest": 2},
364             {"id": _id("clip"), "lesson_id": demo["id"],
365 "audio_asset_id": asset_id, "label": "听写片段", "start_ms": 24000,
366 "end_ms": 38000, "default_speed": 0.8, "loop_suggest": 3},
367             {"id": _id("clip"), "lesson_id": demo["id"],
368 "audio_asset_id": asset_id, "label": "跟读收尾", "start_ms": 38000,
369 "end_ms": 52000, "default_speed": 1.0, "loop_suggest": 1},
370         ]
371     )
372 [padding]
373     def get_user_by_username(self, username: str) -> dict | None:
374         for u in self._users:
375             if u["username"] == username:
376                 return u
377         return None
378 [padding]
379     def list_scenes(self) -> list[dict]:
380         return list(self._scenes)
381 [padding]
382     def get_scene(self, scene_id: str) -> dict | None:
383         for s in self._scenes:
384             if s["id"] == scene_id:
385                 return s
386         return None
387 [padding]
388     def update_scene_capabilities(self, scene_id: str, capabilities: dict) -
389 > dict | None:
390         scene = self.get_scene(scene_id)
391         if not scene:
392             return None
393         merged = dict(scene.get("capabilities") or {})
394         for k in ("force_sync", "student_seek", "student_speed",
395 "student_loop"):
396             if k in capabilities:
397                 merged[k] = bool(capabilities[k])
398         scene["capabilities"] = merged
399         return scene
400 [padding]

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401     def list_devices(self, scene_id: str | None = None) -> list[dict]:
402         if not scene_id:
403             return list(self._devices)
404         return [d for d in self._devices if d["scene_id"] == scene_id]
405     [padding]
406     def list_lessons(self) -> list[dict]:
407         items = []
408         for l in self._lessons:
409             scene = self.get_scene(l["scene_id"])
410             items.append({**l, "scene_name": scene["name"] if scene else
411 l["scene_id"]})
412         return items
413     [padding]
414     def create_lesson(self, title: str, scene_id: str, teacher_id: str,
415 book_unit: str) -> dict:
416         item = {"id": _id("lesson"), "title": title, "scene_id": scene_id,
417 "teacher_id": teacher_id, "book_unit": book_unit, "status": "draft"}
418         self._lessons.append(item)
419         return item
420     [padding]
421     def get_lesson(self, lesson_id: str) -> dict | None:
422         for l in self._lessons:
423             if l["id"] == lesson_id:
424                 return l
425         return None
426     [padding]
427     def list_clips(self, lesson_id: str) -> list[dict]:
428         return [c for c in self._clips if c["lesson_id"] == lesson_id]
429     [padding]
430     def create_clip(
431         self,
432         lesson_id: str,
433         audio_asset_id: str,
434         label: str,
435         start_ms: int,
436         end_ms: int,
437         default_speed: float,
438         loop_suggest: int,
439     ) -> dict:
440         item = {
441             "id": _id("clip"),
442             "lesson_id": lesson_id,
443             "audio_asset_id": audio_asset_id,
444             "label": label,
445             "start_ms": start_ms,
446             "end_ms": end_ms,
447             "default_speed": default_speed,
448             "loop_suggest": loop_suggest,
449         }
450         self._clips.append(item)

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451         return item
452     [padding]
453     def start_session(self, lesson_id: str) -> dict | None:
454         lesson = self.get_lesson(lesson_id)
455         if not lesson:
456             return None
457         lesson["status"] = "running"
458         session = {"id": _id("sess"), "lesson_id": lesson_id, "started_at":
459 _now_ms(), "ended_at": None}
460         self._sessions.append(session)
461         return session
462     [padding]
463     def end_session(self, session_id: str) -> dict | None:
464         sess = next((s for s in self._sessions if s["id"] == session_id),
465 None)
466         if not sess:
467             return None
468         if not sess.get("ended_at"):
469             sess["ended_at"] = _now_ms()
470         lesson = self.get_lesson(sess["lesson_id"])
471         if lesson:
472             lesson["status"] = "ended"
473         return sess
474     [padding]
475     def end_active_session(self, lesson_id: str) -> dict | None:
476         sess = next((s for s in reversed(self._sessions) if s["lesson_id"] ==
477 lesson_id and not s.get("ended_at")), None)
478         if not sess:
479             return None
480         return self.end_session(sess["id"])
481     [padding]
482     def _next_seq(self) -> int:
483         self._seq += 1
484         return self._seq
485     [padding]
486     def add_event(self, session_id: str, actor_user_id: str, target: str,
487 action: str, payload: dict) -> dict | None:
488         sess = next((s for s in self._sessions if s["id"] == session_id),
489 None)
490         if not sess:
491             return None
492         seq = self._next_seq()
493         summary = self._event_summary(action=action, payload=payload)
494         evt = {
495             "seq": seq,
496             "id": _id("evt"),
497             "session_id": session_id,
498             "actor_user_id": actor_user_id,
499             "target": target,
500             "action": action,
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501         "payload": payload,
502         "summary": summary,
503         "created_at": _now_ms(),
504     }
505     self._events.append(evt)
506     return evt
507 [padding]
508     def list_events(self, session_id: str, since: int = 0) -> list[dict]:
509         items = [e for e in self._events if e["session_id"] == session_id
510 and e["seq"] > since]
511         return items[-80:]
512 [padding]
513     def get_session_policy(self, session_id: str) -> dict | None:
514         sess = next((s for s in self._sessions if s["id"] == session_id),
515 None)
516         if not sess:
517             return None
518         lesson = self.get_lesson(sess["lesson_id"])
519         if not lesson:
520             return None
521         scene = self.get_scene(lesson["scene_id"])
522         if not scene:
523             return None
524         return {"scene_id": scene["id"], "scene_name": scene["name"],
525 "capabilities": scene["capabilities"]}
526 [padding]
527     def lesson_stats(self, lesson_id: str) -> dict | None:
528         lesson = self.get_lesson(lesson_id)
529         if not lesson:
530             return None
531         sessions = [s for s in self._sessions if s["lesson_id"] == lesson_id]
532 [padding]
533         events = [e for e in self._events if any(s["id"] == e["session_id"]
534 for s in sessions)]
535         by_action: dict[str, int] = {}
536         for e in events:
537             by_action[e["action"]] = by_action.get(e["action"], 0) + 1
538         return {"lesson": lesson, "session_count": len(sessions),
539 "event_count": len(events), "action_count": by_action}
540 [padding]
541     def _event_summary(self, action: str, payload: dict) -> str:
542         if action == "play":
543             return f"播放片段: {payload.get('label', '')}"
544         if action == "pause":
545             return "暂停播放"
546         if action == "ab_loop":
547             return "A-B 循环"
548         if action == "speed":
549             return f"倍速 {payload.get('speed', 1)}"
550         return action

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551 [padding]
552 [padding]
553 db = InMemoryDB(base_dir=Path(__file__).resolve().parents[2])
554 # apps/_init__.py
555 [padding]
556 # apps/api/_init__.py
557 [padding]
558 # apps/api/data/audio_library.json
559 {"id": "asset_0001", "title": "Unit 2 - Listening Clip 0001",
560 "duration_sec": 19, "level": "A2", "scene_tag": "language_lab",
561 "source_uri": "/static/audio/unit02/clip_0001.mp3"}
562 {"id": "asset_0002", "title": "Unit 3 - Listening Clip 0002",
563 "duration_sec": 20, "level": "B1", "scene_tag": "live", "source_uri":
564 "/static/audio/unit03/clip_0002.mp3"}
565 {"id": "asset_0003", "title": "Unit 4 - Listening Clip 0003",
566 "duration_sec": 21, "level": "B2", "scene_tag": "corner", "source_uri":
567 "/static/audio/unit04/clip_0003.mp3"}
568 {"id": "asset_0004", "title": "Unit 5 - Listening Clip 0004",
569 "duration_sec": 22, "level": "A1", "scene_tag": "selfstudy", "source_uri":
570 "/static/audio/unit05/clip_0004.mp3"}
571 {"id": "asset_0005", "title": "Unit 6 - Listening Clip 0005",
572 "duration_sec": 23, "level": "A2", "scene_tag": "classroom", "source_uri":
573 "/static/audio/unit06/clip_0005.mp3"}
574 {"id": "asset_0006", "title": "Unit 7 - Listening Clip 0006",
575 "duration_sec": 24, "level": "B1", "scene_tag": "language_lab",
576 "source_uri": "/static/audio/unit07/clip_0006.mp3"}
577 {"id": "asset_0007", "title": "Unit 8 - Listening Clip 0007",
578 "duration_sec": 25, "level": "B2", "scene_tag": "live", "source_uri":
579 "/static/audio/unit08/clip_0007.mp3"}
580 {"id": "asset_0008", "title": "Unit 9 - Listening Clip 0008",
581 "duration_sec": 26, "level": "A1", "scene_tag": "corner", "source_uri":
582 "/static/audio/unit09/clip_0008.mp3"}
583 {"id": "asset_0009", "title": "Unit 10 - Listening Clip 0009",
584 "duration_sec": 27, "level": "A2", "scene_tag": "selfstudy", "source_uri":
585 "/static/audio/unit10/clip_0009.mp3"}
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1504 let token = null;
1505 let currentLesson = null;
1506 let currentScene = null;
1507 let currentSession = null;
1508 let currentClip = null;
1509 let since = 0;
1510 [padding]
1511 function el(id){ return document.getElementById(id); }
1512 function fmtTs(ms){ return new Date(ms).toLocaleTimeString(); }
1513 [padding]
1514 function showBanner(text){
1515     const b = el("banner");
1516     if(!b) return;
1517     b.textContent = text;
1518     b.classList.remove("hidden");
1519 }
1520 [padding]
1521 function clearBanner(){
1522     const b = el("banner");
1523     if(!b) return;
1524     b.textContent = "";
1525     b.classList.add("hidden");
1526 }
1527 [padding]
1528 function ensureModal(){
1529     let shell = document.getElementById("modal");
1530     if(shell) return shell;
1531     shell = document.createElement("div");
1532     shell.id = "modal";
1533     shell.className = "modal-shell hidden";
1534     shell.innerHTML = `
1535     <div class="modal-mask"></div>
1536     <div class="modal" id="modalBox">
1537         <div class="modal-title">提示</div>
1538         <div class="modal-body" id="modalBody"></div>
1539         <div class="modal-actions">
1540             <button class="btn" id="modalCancel">取消</button>
1541             <button class="btn primary" id="modalOk">确定</button>
1542         </div>
1543     </div>`;
1544     document.body.appendChild(shell);
1545     const close = () => shell.classList.add("hidden");
1546     el("modalCancel").onclick = close;
1547     el("modalOk").onclick = close;
1548     window._rkModal = {
1549         closeModal: close,
1550         open: (text) => {
```

```
1551     el("modalBody").textContent = text;
1552     shell.classList.remove("hidden");
1553 }
1554 };
1555 return shell;
1556 }
1557 [padding]
1558 function showTip(text){
1559     ensureModal();
1560     window.__rkModal.open(text);
1561 }
1562 [padding]
1563 function showTipKind(kind, text){
1564     ensureModal();
1565     const shell = document.getElementById("modal");
1566     if(shell){
1567         shell.classList.remove("kind-danger","kind-warning");
1568         if(kind === "danger") shell.classList.add("kind-danger");
1569         if(kind === "warning") shell.classList.add("kind-warning");
1570     }
1571     const box = el("modalBox");
1572     if(box){
1573         box.classList.remove("danger","warning");
1574         if(kind) box.classList.add(kind);
1575     }
1576     window.__rkModal.open(text);
1577 }
1578 [padding]
1579 async function api(path, opts={}){
1580     const res = await fetch(API + path, {
1581         ...opts,
1582         headers: {
1583             "Content-Type":"application/json",
1584             ...(opts.headers||{}),
1585             ...(token ? {"Authorization": token} : {}),
1586         }
1587     });
1588     if(!res.ok){
1589         const t = await res.text();
1590         throw new Error(t || ("HTTP " + res.status));
1591     }
1592     return res.json();
1593 }
1594 [padding]
1595 async function refreshScenes(){
1596     const data = await api("/scenes");
1597     const sel = el("sceneSelect");
1598     sel.innerHTML = "";
1599     data.items.forEach(s=>{
1600         const opt = document.createElement("option");
```



```
1601     opt.value = s.id;
1602     opt.textContent = s.name;
1603     sel.appendChild(opt);
1604   });
1605   currentScene = data.items[0] || null;
1606   if(currentScene) sel.value = currentScene.id;
1607   sel.onchange = async () => {
1608     currentScene = data.items.find(x=>x.id===sel.value) || null;
1609     await refreshDevices();
1610   };
1611   await refreshDevices();
1612 }
1613 [padding]
1614 async function refreshDevices(){
1615   const data = await api("/devices" + (currentScene ? ("?scene_id=" +
1616   encodeURIComponent(currentScene.id)) : ""));
1617   const box = el("devices");
1618   box.innerHTML = "";
1619   data.items.forEach(d=>{
1620     const row = document.createElement("div");
1621     row.className = "device";
1622     row.innerHTML = `<div><div style="font-weight:600">${d.name}</div><div
1623   style="color:var(--muted);font-size:12px">${d.type}</div></div>`;
1624     const b = document.createElement("div");
1625     b.className = "badge " + (d.online ? "on" : "off");
1626     b.textContent = d.online ? "在线" : "离线";
1627     row.appendChild(b);
1628     box.appendChild(row);
1629   });
1630 }
1631 [padding]
1632 async function refreshLessons(){
1633   const data = await api("/lessons");
1634   const sel = el("lessonSelect");
1635   sel.innerHTML = "";
1636   data.items.forEach(l=>{
1637     const opt = document.createElement("option");
1638     opt.value = l.id;
1639     opt.textContent = l.title + " (" + l.scene_name + ") ";
1640     sel.appendChild(opt);
1641   });
1642   currentLesson = data.items[0] || null;
1643   if(currentLesson) sel.value = currentLesson.id;
1644   sel.onchange = async () => {
1645     currentLesson = data.items.find(x=>x.id===sel.value) || null;
1646     await refreshClips();
1647   };
1648   await refreshClips();
1649 }
1650 [padding]
```

```

1651 async function refreshClips(){
1652     const box = el("clips");
1653     box.innerHTML = "";
1654     currentClip = null;
1655     if(!currentLesson) return;
1656     const data = await api(`/lessons/${currentLesson.id}/clips`);
1657     data.items.forEach(c=>{
1658         const d = document.createElement("div");
1659         d.className = "clip";
1660         d.innerHTML = `<div class="t">${c.label}</div><div class="m">${(c.end_ms-
1661 c.start_ms)/1000}s · 默认倍速 ${c.default_speed}x</div>`;
1662         d.onclick = () => {
1663             currentClip = c;
1664             document.querySelectorAll(".clip").forEach(x=>
1665 x.classList.remove("active"));
1666             d.classList.add("active");
1667             showBanner(`已选择片段: ${c.label}`);
1668         };
1669         box.appendChild(d);
1670     });
1671 }
1672 [padding]
1673 function renderPolicy(policy){
1674     if(!policy){ el("policy").textContent = ""; return; }
1675     const p = policy.capabilities;
1676     el("policy").textContent = `场景策略: 强制同步=${p.force_sync ? "是":"否"}
1677 ; 学生跳转=${p.student_seek ? "允许":"禁止"}; 倍速=${p.student_speed ?
1678 "允许":"禁止"}; 循环=${p.student_loop ? "允许":"禁止"}`;
1679 }
1680 [padding]
1681 function pushEvent(evt){
1682     const tl = el("timeline");
1683     const row = document.createElement("div");
1684     row.className = "evt";
1685     row.innerHTML = `<div class="dot"></div><div class="c"><div class=
1686 "ts">${fmtTs(evt.created_at)}</div><div class="msg">${evt.action} ·
1687 ${evt.summary}</div></div>`;
1688     tl.prepend(row);
1689 }
1690 [padding]
1691 async function pollEvents(){
1692     if(!currentSession) return;
1693     const data = await api(`/sessions/${currentSession.id}/events?since=
1694 ${since}`);
1695     data.items.forEach(e=>{
1696         since = Math.max(since, e.seq);
1697         pushEvent(e);
1698     });
1699     setTimeout(pollEvents, 1200);
1700 }

```

```

1701 [padding]
1702 async function ensureSessionPolicy(){
1703   if(!currentSession) return;
1704   const data = await api(`/sessions/${currentSession.id}/policy`);
1705   renderPolicy(data.policy);
1706 }
1707 [padding]
1708 async function main(){
1709   await refreshScenes();
1710   await refreshLessons();
1711   clearBanner();
1712 [padding]
1713   window.__shotTag = (text) => {
1714     const t = String(text || "").trim();
1715     const b = el("banner");
1716     if(b){
1717       b.className = b.className.replace(/^btag-\d{2}\b/g, "").trim();
1718       const m = t.match(/^截图(\d{2})/);
1719       if(m){
1720         b.classList.add(`tag-${m[1]}`);
1721       }
1722     }
1723     showBanner(t);
1724   };
1725 [padding]
1726   el("btnLogin").onclick = async () => {
1727     const u = el("u").value.trim();
1728     const p = el("p").value.trim();
1729     const data = await api("/login", {method:"POST",
1730 body:JSON.stringify({username:u, password:p})});
1731     token = data.token;
1732     currentUser = data.user;
1733     el("who").textContent = `已登录: ${currentUser.name} (${currentUser.role}
1734 )`;
1735     showBanner(`登录成功: ${currentUser.role} (用于截图与演示)`);
1736   };
1737 [padding]
1738   el("btnAddLesson").onclick = async () => {
1739     if(!currentUser || currentUser.role!=="teacher"){
1740       alert("请先用 teacher 登录创建课时");
1741       return;
1742     }
1743     const title = "Unit " + (Math.floor(Math.random()*6)+1) + " 听力训练";
1744     const sceneId = el("sceneSelect").value;
1745     const data = await api("/lessons", {method:"POST",
1746 body:JSON.stringify({title, scene_id:sceneId, teacher_id: currentUser.id,
1747 book_unit:"Demo"})});
1748     currentLesson = data.item;
1749     await refreshLessons();
1750     showBanner(`已新增课时: ${currentLesson.title}`);

```

```
1751     };
1752 [padding]
1753     el("btnEditLesson").onclick = async () => {
1754         if(!currentLesson){ alert("请选择课时"); return; }
1755         showTip("演示版： 编辑课时仅展示按钮与流程入口");
1756     };
1757 [padding]
1758     el("btnDeleteLesson").onclick = async () => {
1759         if(!currentLesson){ alert("请选择课时"); return; }
1760         showTipKind("danger","演示版： 删除课时确认（演示）");
1761     };
1762 [padding]
1763     el("btnStart").onclick = async () => {
1764         if(!currentUser || currentUser.role!=="teacher"){
1765             alert("请先用 teacher 登录开始上课");
1766             return;
1767         }
1768         if(!currentLesson){ alert("请选择课时"); return; }
1769         const data = await api(`/lessons/${currentLesson.id}/start`,
1770 {method:"POST", body:JSON.stringify({}}));
1771         currentSession = data.session;
1772         since = 0;
1773         el("timeline").innerHTML = "";
1774         await ensureSessionPolicy();
1775         pollEvents();
1776         showBanner(`课堂已开始： session=${currentSession.id}`);
1777     };
1778 [padding]
1779     function requireClip(){
1780         if(!currentClip){ alert("请选择片段"); return false; }
1781         if(!currentSession){ alert("请先开始上课"); return false; }
1782         if(!currentUser){ alert("请先登录"); return false; }
1783         return true;
1784     }
1785 [padding]
1786     el("btnPlay").onclick = async () => {
1787         if(!requireClip()) return;
1788         await api(`/sessions/${currentSession.id}/event`, {method:"POST",
1789 body:JSON.stringify({
1790     action:"play", actor_user_id: currentUser.id, target:"all_students",
1791     payload:{clip_id: currentClip.id, label: currentClip.label}
1792     }));
1793     };
1794     el("btnPause").onclick = async () => {
1795         if(!currentSession || !currentUser) return;
1796         await api(`/sessions/${currentSession.id}/event`, {method:"POST",
1797 body:JSON.stringify({action:"pause", actor_user_id: currentUser.id,
1798 target:"all_students", payload:{}}));
1799     };
1800     el("btnLoop").onclick = async () => {
```

```
1801     if(!requireClip()) return;
1802     await api('/sessions/${currentSession.id}/event', {method:"POST",
1803 body:JSON.stringify({action:"ab_loop", actor_user_id: currentUser.id,
1804 target:"all_students", payload:{clip_id: currentClip.id, a_ms:
1805 currentClip.start_ms, b_ms: currentClip.end_ms}}));
1806 };
1807 el("btnSpeed").onclick = async () => {
1808     if(!requireClip()) return;
1809     await api('/sessions/${currentSession.id}/event', {method:"POST",
1810 body:JSON.stringify({action:"speed", actor_user_id: currentUser.id,
1811 target:"all_students", payload:{clip_id: currentClip.id, speed: 1.2}}));
1812 };
1813 [padding]
1814 el("btnAddClip").onclick = async () => {
1815     if(!currentLesson){ alert("请选择课时"); return; }
1816     showTip("演示版： 新增片段入口（可扩展为表单）");
1817 };
1818 el("btnEditClip").onclick = async () => {
1819     if(!currentClip){ alert("请选择片段"); return; }
1820     showTip("演示版： 编辑片段入口（可扩展为表单）");
1821 };
1822 el("btnDeleteClip").onclick = async () => {
1823     if(!currentClip){ alert("请选择片段"); return; }
1824     showTipKind("warning","演示版： 删除片段确认（演示）");
1825 };
1826 }
1827 [padding]
1828 main().catch(err=>{
1829     console.error(err);
1830     alert("页面初始化失败： " + err.message);
1831 });
1832 # apps/web/index.html
1833 <!doctype html>
1834 <html lang="zh-CN">
1835 <head>
1836     <meta charset="utf-8" />
1837     <meta name="viewport" content="width=device-width, initial-scale=1" />
1838     <title>多场景英语教学音频播放与管控软件</title>
1839     <link rel="stylesheet" href="/styles.css" />
1840 </head>
1841 <body>
1842     <div class="app">
1843         <header class="topbar">
1844             <div class="brand">
1845                 <div class="logo"></div>
1846                 <div>
1847                     <div class="brand-title">多场景英语教学音频播放与管控软件</div>
1848                     <div class="brand-sub">教师端 / 学生端 / 管理端</div>
1849                 </div>
1850             </div>
```

```
1851     <nav class="nav">
1852         <a class="nav-link" href="/teacher.html">教师端</a>
1853         <a class="nav-link" href="/student.html">学生端</a>
1854         <a class="nav-link" href="/admin.html">管理端</a>
1855     </nav>
1856 </header>
1857 [padding]
1858 <main class="card landing">
1859     <h1>入口</h1>
1860     <p class="muted">选择一个端进入系统。演示版不使用复杂构建工具，页面
1861 为静态多页。</p>
1862     <div class="landing-actions">
1863         <a class="btn primary" href="/teacher.html">进入教师端</a>
1864         <a class="btn" href="/student.html">进入学生端</a>
1865         <a class="btn" href="/admin.html">进入管理端</a>
1866     </div>
1867 </main>
1868 </div>
1869 </body>
1870 </html>
1871 [padding]
1872 # apps/web/shared.js
1873 const API = "/api";
1874 [padding]
1875 export function el(id) {
1876     return document.getElementById(id);
1877 }
1878 [padding]
1879 export function qs(sel) {
1880     return document.querySelector(sel);
1881 }
1882 [padding]
1883 export function qsa(sel) {
1884     return Array.from(document.querySelectorAll(sel));
1885 }
1886 [padding]
1887 export function fmtTs(ms) {
1888     return new Date(ms).toLocaleTimeString();
1889 }
1890 [padding]
1891 export function getAuth() {
1892     const raw = localStorage.getItem("demo_auth");
1893     if (!raw) return { token: null, user: null };
1894     try {
1895         const parsed = JSON.parse(raw);
1896         return { token: parsed.token || null, user: parsed.user || null };
1897     } catch {
1898         return { token: null, user: null };
1899     }
1900 }
```

```
1901 [padding]
1902 export function setAuth(token, user) {
1903   localStorage.setItem("demo_auth", JSON.stringify({ token, user }));
1904 }
1905 [padding]
1906 export function clearAuth() {
1907   localStorage.removeItem("demo_auth");
1908 }
1909 [padding]
1910 export async function api(path, opts = {}) {
1911   const { token } = getAuth();
1912   const res = await fetch(API + path, {
1913     ...opts,
1914     headers: {
1915       "Content-Type": "application/json",
1916       ...(opts.headers || {}),
1917       ...(token ? { Authorization: token } : {}),
1918     },
1919   });
1920   if (!res.ok) {
1921     const t = await res.text();
1922     throw new Error(t || `HTTP ${res.status}`);
1923   }
1924   return res.json();
1925 }
1926 [padding]
1927 export function ensureModal() {
1928   let shell = document.getElementById("modal");
1929   if (shell) return shell;
1930   shell = document.createElement("div");
1931   shell.id = "modal";
1932   shell.className = "modal-shell hidden";
1933   shell.innerHTML = `
1934     <div class="modal-mask"></div>
1935     <div class="modal" id="modalBox">
1936       <div class="modal-title" id="modalTitle">提示</div>
1937       <div class="modal-body" id="modalBody"></div>
1938       <div class="modal-actions">
1939         <button class="btn" id="modalCancel">取消</button>
1940         <button class="btn primary" id="modalOk">确定</button>
1941       </div>
1942     </div>`;
1943   document.body.appendChild(shell);
1944 [padding]
1945   const close = () => shell.classList.add("hidden");
1946   el("modalCancel").onclick = close;
1947   el("modalOk").onclick = close;
1948 [padding]
1949   window.__rkModal = {
1950     closeModal: close,
```

```
1951     open: (text) => {
1952         el("modalBody").textContent = text;
1953         shell.classList.remove("hidden");
1954     },
1955 };
1956 return shell;
1957 }
1958 [padding]
1959 export function showTipKind(kind, text) {
1960     ensureModal();
1961     const shell = document.getElementById("modal");
1962     if (shell) {
1963         shell.classList.remove("kind-danger", "kind-warning");
1964         if (kind === "danger") shell.classList.add("kind-danger");
1965         if (kind === "warning") shell.classList.add("kind-warning");
1966     }
1967     const box = el("modalBox");
1968     const title = el("modalTitle");
1969     if (box) {
1970         box.classList.remove("danger", "warning");
1971         if (kind) box.classList.add(kind);
1972     }
1973     if (title) {
1974         title.textContent = kind === "danger" ? "确认删除" : kind === "warning"
1975 ? "确认操作" : "提示";
1976     }
1977     window.__rkModal.open(text);
1978 }
1979 [padding]
1980 export function showBanner(text) {
1981     const b = el("banner");
1982     if (!b) return;
1983     b.textContent = text;
1984     b.classList.remove("hidden");
1985 }
1986 [padding]
1987 export function clearBanner() {
1988     const b = el("banner");
1989     if (!b) return;
1990     b.textContent = "";
1991     b.classList.add("hidden");
1992 }
1993 [padding]
1994 export function installShotTag() {
1995     window.__shotTag = (text) => {
1996         const t = String(text || "").trim();
1997         const m = t.match(/^(截图(\d{2}))/);
1998         if (m) {
1999             document.body.dataset.shot = m[1];
2000         }
2001     }
2002 }
```



```

2001     const b = el("banner");
2002     if (b) {
2003         b.className = b.className.replace(/\\btag-\\d{2}\\b/g, "").trim();
2004         if (m) {
2005             b.classList.add(`tag-${m[1]}`);
2006             const pal = {
2007                 "01": { bg: "#ecfeff", primary: "#0891b2" },
2008                 "02": { bg: "#eff6ff", primary: "#2563eb" },
2009                 "03": { bg: "#f0fdf4", primary: "#16a34a" },
2010                 "04": { bg: "#fff7ed", primary: "#ea580c" },
2011                 "05": { bg: "#fdf2f8", primary: "#db2777" },
2012                 "06": { bg: "#f5f3ff", primary: "#7c3aed" },
2013                 "07": { bg: "#fef2f2", primary: "#dc2626" },
2014                 "08": { bg: "#fffbeb", primary: "#d97706" },
2015                 "09": { bg: "#f8faff", primary: "#334155" },
2016                 "10": { bg: "#fff5f9", primary: "#f172a" },
2017                 "11": { bg: "#eef2ff", primary: "#4f46e5" },
2018                 "12": { bg: "#ecfcb", primary: "#65a30d" },
2019             }[m[1]];
2020             if (pal) {
2021                 document.documentElement.style.setProperty("--bg", pal.bg);
2022                 document.documentElement.style.setProperty("--primary",
2023                 pal.primary);
2024             }
2025         }
2026     }
2027     showBanner(t);
2028 };
2029 }
2030 # apps/web/student.html
2031 <!doctype html>
2032 <html lang="zh-CN">
2033 <head>
2034     <meta charset="utf-8" />
2035     <meta name="viewport" content="width=device-width, initial-scale=1" />
2036     <title>学生端 - 多场景英语教学音频播放与管控软件</title>
2037     <link rel="stylesheet" href="/styles.css" />
2038 </head>
2039 <body>
2040     <div class="app">
2041         <header class="topbar">
2042             <div class="brand">
2043                 <div class="logo"></div>
2044                 <div>
2045                     <div class="brand-title">学生端</div>
2046                     <div class="brand-sub">事件跟随 / 场景策略</div>
2047                 </div>
2048             </div>
2049             <nav class="nav">
2050                 <a class="nav-link" href="/index.html">入口</a>

```

```

2051     <a class="nav-link" href="/teacher.html">教师端</a>
2052     <a class="nav-link active" href="/student.html">学生端</a>
2053     <a class="nav-link" href="/admin.html">管理端</a>
2054 </nav>
2055     <div class="who" id="who">未登录</div>
2056 </header>
2057 [padding]
2058     <main class="grid">
2059         <section class="card span3 banner hidden" id="banner"></section>
2060 [padding]
2061         <section class="card">
2062             <h2>登录</h2>
2063             <div class="row">
2064                 <label>账号</label>
2065                 <input id="u" placeholder="student" />
2066             </div>
2067             <div class="row">
2068                 <label>密码</label>
2069                 <input id="p" type="password" placeholder="demo" />
2070             </div>
2071             <button id="btnLogin" class="btn primary">登录</button>
2072             <div class="hint">演示账号： student； 密码： demo</div>
2073         </section>
2074 [padding]
2075         <section class="card span2">
2076             <h2>课堂会话</h2>
2077             <div class="row">
2078                 <label>Session</label>
2079                 <input id="sessionId" placeholder="输入 session
2080 id（由教师端开始上课生成）" />
2081                 <button id="btnFollow" class="btn primary">开始跟随</button>
2082             </div>
2083             <div class="policy" id="policy"></div>
2084             <div class="timeline" id="timeline"></div>
2085         </section>
2086     </main>
2087 </div>
2088     <script type="module" src="/student.js"></script>
2089 </body>
2090 </html>
2091 [padding]
2092 # apps/web/student.js
2093 import { api, clearBanner, el, fmtTs, getAuth, installShotTag, setAuth,
2094 showBanner, showTipKind } from "/shared.js";
2095 [padding]
2096 let currentSessionId = null;
2097 let since = 0;
2098 [padding]
2099 function renderPolicy(policy) {
2100     if (!policy) {

```

```

2101     el("policy").textContent = "";
2102     return;
2103 }
2104 const p = policy.capabilities;
2105 el("policy").textContent = `场景策略：强制同步=${p.force_sync ? "是" :
2106 "否"}；学生跳转=${p.student_seek ? "允许" : "禁止"}；倍速=${p.student_speed
2107 ? "允许" : "禁止"}；循环=${p.student_loop ? "允许" : "禁止"}`;
2108 }
2109 [padding]
2110 function pushEvent(evt) {
2111     const tl = el("timeline");
2112     const row = document.createElement("div");
2113     row.className = "evt";
2114     row.innerHTML = `<div class="dot"></div><div class="c"><div class=
2115 "ts">${fmtTs(evt.created_at)}</div><div class="msg">${evt.action} ·
2116 ${evt.summary}</div></div>`;
2117     tl.prepend(row);
2118 }
2119 [padding]
2120 async function poll() {
2121     if (!currentSessionId) return;
2122     const data = await api(`/sessions/${currentSessionId}/events?since=${since}
2123 `);
2124     data.items.forEach((e) => {
2125         since = Math.max(since, e.seq);
2126         pushEvent(e);
2127     });
2128     setTimeout(poll, 900);
2129 }
2130 [padding]
2131 async function follow(sessionId) {
2132     currentSessionId = sessionId;
2133     since = 0;
2134     el("timeline").innerHTML = "";
2135     const pol = await api(`/sessions/${currentSessionId}/policy`);
2136     renderPolicy(pol.policy);
2137     showBanner(`开始跟随： ${pol.policy.scene_name}`);
2138     poll();
2139 }
2140 [padding]
2141 async function main() {
2142     installShotTag();
2143     clearBanner();
2144     const { user } = getAuth();
2145     if (user) el("who").textContent = `已登录： ${user.name}（${user.role}）`;
2146 [padding]
2147     el("btnLogin").onclick = async () => {
2148         const u = el("u").value.trim();
2149         const p = el("p").value.trim();
2150         const data = await api("/login", { method: "POST", body:

```

```
2151 JSON.stringify({ username: u, password: p }));
2152     setAuth(data.token, data.user);
2153     el("who").textContent = `已登录: ${data.user.name} (${data.user.role})`;
2154 [padding]
2155     showBanner(`登录成功: ${data.user.role}`);
2156 };
2157 [padding]
2158 el("btnFollow").onclick = async () => {
2159     const sid = el("sessionId").value.trim();
2160     if (!sid) return;
2161     await follow(sid);
2162 };
2163 }
2164 [padding]
2165 main().catch((err) => {
2166     console.error(err);
2167     showTipKind("warning", "页面初始化失败: " + err.message);
2168 });
2169 [padding]
2170 # apps/web/styles.css
2171 :root{
2172     --bg:#f5f7fb;
2173     --card:#ffffff;
2174     --text:#0f172a;
2175     --muted:#64748b;
2176     --primary:#2563eb;
2177     --primary-2:#1d4ed8;
2178     --border:#e2e8f0;
2179 }
2180 *{box-sizing:border-box;}
2181 body{
2182     margin:0;
2183     font-family: ui-sans-serif, system-ui, -apple-system, Segoe UI, Roboto,
2184     Arial, "PingFang SC", "Microsoft YaHei", sans-serif;
2185     color:var(--text);
2186     background:var(--bg);
2187 }
2188 body::before{
2189     content:"";
2190     position:fixed;
2191     width:220px;
2192     height:220px;
2193     left:-9999px;
2194     top:-9999px;
2195     background:#000;
2196     opacity:.12;
2197     border-radius:26px;
2198     z-index:0;
2199 }
2200 .app{position:relative; z-index:1;}
```

```
2201 body[data-shot="03"]::before { left:22px; top:22px; background:#16a34a;
2202 opacity:.22; }
2203 body[data-shot="04"]::before { right:22px; left:auto; top:22px;
2204 background:#ea580c; opacity:.22; }
2205 body[data-shot="06"]::before { left:22px; top:auto; bottom:22px;
2206 background:#7c3a6d; opacity:.22; }
2207 body[data-shot="07"]::before { right:22px; left:auto; top:auto; bottom:22px;
2208 background:#dc2626; opacity:.22; }
2209 .app {max-width:1200px;margin:0 auto;padding:16px;}
2210 .topbar {
2211   display:flex;align-items:center;justify-content:space-between;
2212   padding:14px 16px;border:1px solid var(--border);background:var(--card);
2213   border-radius:14px;
2214 }
2215 .brand {display:flex;gap:12px;align-items:center;}
2216 .logo {width:38px;height:38px;border-radius:10px;background:linear-
2217 gradient(135deg,var(--primary),#60a5fa);}
2218 .brand-title {font-weight:700;}
2219 .brand-sub {color:var(--muted);font-size:12px;margin-top:2px;}
2220 .who {color:var(--muted);font-size:13px;}
2221 .nav {display:flex;gap:10px;align-items:center;}
2222 .nav-link {
2223   color:var(--muted);
2224   text-decoration:none;
2225   font-size:13px;
2226   padding:8px 10px;
2227   border-radius:10px;
2228   border:1px solid transparent;
2229 }
2230 .nav-link:hover {border-color:var(--border);background:#fff;}
2231 .nav-link.active {color:var(--text);border-color:#bfdbfe;background:#eff6ff;}
2232 .grid {display:grid;grid-template-columns:repeat(3,1fr);gap:12px;margin-
2233 top:12px;}
2234 .card {
2235   background:var(--card);border:1px solid var(--border);border-radius:14px;
2236   padding:14px 14px 12px;
2237   box-shadow:0 1px 0 rgba(15,23,42,.03);
2238 }
2239 .span2 {grid-column:span 2;}
2240 h2 {margin:0 0 10px;font-size:16px;}
2241 h1 {margin:0 0 10px;font-size:18px;}
2242 .landing {padding:18px;}
2243 .muted {color:var(--muted);}
2244 .landing-actions {display:flex;gap:10px;flex-wrap:wrap;margin-top:10px;}
2245 .chk {display:flex;gap:8px;align-items:center;color:var(--text);font-
2246 size:13px;margin:8px 0;}
2247 .policy-editor {display:grid;grid-template-columns:repeat(2, minmax(0,1fr));
2248 gap:8px;padding:6px 0;}
2249 .row {display:flex;gap:10px;align-items:center;margin:8px 0;}
2250 label {width:56px;color:var(--muted);font-size:13px;}
```

```

2251 input,select{
2252     flex:1;padding:10px 10px;border-radius:10px;border:1px solid var(--border);
2253     background:#fff;
2254 }
2255 .btn{
2256     padding:10px 12px;border-radius:10px;border:1px solid var(--border);
2257     background:#fff;cursor:pointer;
2258 }
2259 .btn.primary{background:var(--primary);border-color:var(--primary);
2260     color:#fff;}
2261 .btn.primary:hover{background:var(--primary-2);}
2262 .hint{color:var(--muted);font-size:12px;margin-top:6px;}
2263 .devices{display:flex;flex-direction:column;gap:8px;}
2264 .device{
2265     display:flex;justify-content:space-between;align-items:center;
2266     padding:10px;border:1px solid var(--border);border-radius:12px;
2267     background:#fff;
2268 }
2269 .badge{font-size:12px;padding:3px 8px;border-radius:999px;border:1px solid
2270     var(--border);color:var(--muted);}
2271 .badge.on{border-color:#86efac;color:#166534;background:#dcfce7;}
2272 .badge.off{border-color:#fecaca;color:#991b1b;background:#fee2e2;}
2273 .clips{display:grid;grid-template-columns:repeat(2,1fr);gap:8px;}
2274 .clip{padding:10px;border:1px solid var(--border);border-radius:12px;
2275     background:#fff;cursor:pointer;}
2276 .clip.active{border-color:#93c5fd;background:#eff6ff;}
2277 .clip .t{font-weight:600;font-size:13px;}
2278 .clip .m{color:var(--muted);font-size:12px;margin-top:4px;}
2279 .controls{display:flex;gap:8px;flex-wrap:wrap;margin-bottom:8px;}
2280 .policy{color:var(--muted);font-size:12px;margin:6px 0 10px;}
2281 .timeline{border-top:1px dashed var(--border);padding-top:10px;display:flex;
2282     flex-direction:column;gap:8px;}
2283 .evt{display:flex;gap:10px;align-items:flex-start;}
2284 .dot{width:10px;height:10px;border-radius:50%;background:#60a5fa;margin-
2285     top:4px;}
2286 .evt .c{flex:1;}
2287 .evt .ts{color:var(--muted);font-size:12px;}
2288 .evt .msg{font-size:13px;margin-top:2px;}
2289 .span3{grid-column:span 3;}
2290 .banner{
2291     border-style:solid;
2292     border-color:#bfdbfe;
2293     background:#eff6ff;
2294     color:#1e3a8a;
2295     font-weight:700;
2296     font-size:16px;
2297     padding:14px 14px;
2298 }
2299 .banner.hidden{display:none;}
2300 [padding]

```

```

2301 .banner.tag-01{ background:#ecfeff; border-color:#67e8f9; color:#155e75; }
2302 .banner.tag-02{ background:#eff6ff; border-color:#93c5fd; color:#1e3a8a; }
2303 .banner.tag-03{ background:#f0fdf4; border-color:#86efac; color:#166534; }
2304 .banner.tag-04{ background:#fff7ed; border-color:#fdb74; color:#9a3412; }
2305 .banner.tag-05{ background:#fdf2f8; border-color:#f9a8d4; color:#9d174d; }
2306 .banner.tag-06{ background:#f5f3ff; border-color:#c4b5fd; color:#5b21b6; }
2307 .banner.tag-07{ background:#fef2f2; border-color:#fca5a5; color:#991b1b; }
2308 .banner.tag-08{ background:#fffbef; border-color:#fcd34d; color:#92400e; }
2309 .banner.tag-09{ background:#f8faff; border-color:#cbd5e1; color:#0f172a; }
2310 .banner.tag-10{ background:#f1f5f9; border-color:#94a3b8; color:#0f172a; }
2311 [padding]
2312 .modal-shell{position:fixed;inset:0;display:flex;align-items:center;justify-
2313 content:center;z-index:50;}
2314 .modal-shell.hidden{display:none;}
2315 .modal-mask{position:absolute;inset:0;background:rgba(15,23,42,.38);}
2316 .modal-shell.kind-danger .modal-mask{background:rgba(127,29,29,.45);}
2317 .modal-shell.kind-warning .modal-mask{background:rgba(146,64,14,.40);}
2318 .modal{
2319   position:relative; width:min(520px, calc(100vw - 32px));
2320   background:#fff;border:1px solid var(--border);border-radius:14px;
2321   padding:12px 12px 10px; box-shadow:0 18px 50px rgba(2,6,23,.22);
2322 }
2323 .modal.danger{ border-color:#fecaca; }
2324 .modal.warning{ border-color:#fde68a; }
2325 .modal .modal-title{ padding:6px 10px; border-radius:10px;
2326 background:#f1f5f9; }
2327 .modal.danger .modal-title{ background:#fee2e2; color:#991b1b; }
2328 .modal.warning .modal-title{ background:#fef3c7; color:#92400e; }
2329 .modal-title{font-weight:700;margin-bottom:8px;}
2330 .modal-body{color:var(--text);line-height:1.6;min-height:44px;}
2331 .modal-actions{display:flex;justify-content:flex-end;gap:8px;margin-top:10px;
2332 }
2333 # apps/web/teacher.html
2334 <!doctype html>
2335 <html lang="zh-CN">
2336 <head>
2337   <meta charset="utf-8" />
2338   <meta name="viewport" content="width=device-width, initial-scale=1" />
2339   <title>教师端 - 多场景英语教学音频播放与管控软件</title>
2340   <link rel="stylesheet" href="/.styles.css" />
2341 </head>
2342 <body>
2343   <div class="app">
2344     <header class="topbar">
2345       <div class="brand">
2346         <div class="logo"></div>
2347         <div>
2348           <div class="brand-title">教师端</div>
2349           <div class="brand-sub">课时 / 片段 / 控制台 / 时间轴</div>
2350         </div>

```

```
2351     </div>
2352     <nav class="nav">
2353         <a class="nav-link" href="/index.html">入口</a>
2354         <a class="nav-link active" href="/teacher.html">教师端</a>
2355         <a class="nav-link" href="/student.html">学生端</a>
2356         <a class="nav-link" href="/admin.html">管理端</a>
2357     </nav>
2358     <div class="who" id="who">未登录</div>
2359 </header>
2360 [padding]
2361     <main class="grid">
2362         <section class="card span3 banner hidden" id="banner"></section>
2363 [padding]
2364     <section class="card">
2365         <h2>登录</h2>
2366         <div class="row">
2367             <label>账号</label>
2368             <input id="u" placeholder="teacher" />
2369         </div>
2370         <div class="row">
2371             <label>密码</label>
2372             <input id="p" type="password" placeholder="demo" />
2373         </div>
2374         <button id="btnLogin" class="btn primary">登录</button>
2375         <div class="hint">演示账号：teacher；密码：demo</div>
2376     </section>
2377 [padding]
2378     <section class="card">
2379         <h2>场景与设备</h2>
2380         <div class="row">
2381             <label>场景</label>
2382             <select id="sceneSelect"></select>
2383         </div>
2384         <div class="devices" id="devices"></div>
2385     </section>
2386 [padding]
2387     <section class="card span2">
2388         <h2>课时与片段</h2>
2389         <div class="row">
2390             <label>课时</label>
2391             <select id="lessonSelect"></select>
2392             <button id="btnAddLesson" class="btn">新增</button>
2393             <button id="btnEditLesson" class="btn">编辑</button>
2394             <button id="btnDeleteLesson" class="btn">删除</button>
2395             <button id="btnStart" class="btn primary">开始上课</button>
2396         </div>
2397         <div class="clips" id="clips"></div>
2398     </section>
2399 [padding]
2400     <section class="card span2">
```



```

2401     <h2>课堂控制台</h2>
2402     <div class="controls">
2403         <button id="btnPlay" class="btn primary">播放</button>
2404         <button id="btnPause" class="btn">暂停</button>
2405         <button id="btnLoop" class="btn">A-B 循环</button>
2406         <button id="btnSpeed" class="btn">倍速 1.2x</button>
2407         <button id="btnEnd" class="btn">结束上课</button>
2408     </div>
2409     <div class="controls">
2410         <button id="btnAddClip" class="btn">新增</button>
2411         <button id="btnEditClip" class="btn">编辑</button>
2412         <button id="btnDeleteClip" class="btn">删除</button>
2413     </div>
2414     <div class="policy" id="policy"></div>
2415     <div class="timeline" id="timeline"></div>
2416 </section>
2417 </main>
2418 </div>
2419 <script type="module" src="/teacher.js"></script>
2420 </body>
2421 </html>
2422 [padding]
2423 # apps/web/teacher.js
2424 import { api, clearBanner, clearAuth, el, getAuth, installShotTag, qsa,
2425 setAuth, showBanner, showTipKind, fmtTs } from "/shared.js";
2426 [padding]
2427 let currentLesson = null;
2428 let currentScene = null;
2429 let currentSession = null;
2430 let currentClip = null;
2431 let since = 0;
2432 [padding]
2433 async function refreshScenes() {
2434     const data = await api("/scenes");
2435     const sel = el("sceneSelect");
2436     sel.innerHTML = "";
2437     data.items.forEach((s) => {
2438         const opt = document.createElement("option");
2439         opt.value = s.id;
2440         opt.textContent = s.name;
2441         sel.appendChild(opt);
2442     });
2443     currentScene = data.items[0] || null;
2444     if (currentScene) sel.value = currentScene.id;
2445     sel.onchange = async () => {
2446         currentScene = data.items.find((x) => x.id === sel.value) || null;
2447         showBanner(`已切换场景: ${currentScene ? currentScene.name : "-"}`);
2448         await refreshDevices();
2449     };
2450     await refreshDevices();

```

```

2451 }
2452 [padding]
2453 async function refreshDevices() {
2454     const data = await api("/devices" + (currentScene ? "?scene_id="
2455     ${encodeURIComponent(currentScene.id)}' : ""));
2456     const box = el("devices");
2457     box.innerHTML = "";
2458     data.items.forEach((d) => {
2459         const row = document.createElement("div");
2460         row.className = "device";
2461         row.innerHTML = `<div><div style="font-weight:600">${d.name}</div><div
2462 style="color:var(--muted);font-size:12px">${d.type}</div></div>`;
2463         const b = document.createElement("div");
2464         b.className = "badge " + (d.online ? "on" : "off");
2465         b.textContent = d.online ? "在线" : "离线";
2466         row.appendChild(b);
2467         box.appendChild(row);
2468     });
2469 }
2470 [padding]
2471 async function refreshLessons() {
2472     const data = await api("/lessons");
2473     const sel = el("lessonSelect");
2474     sel.innerHTML = "";
2475     data.items.forEach((l) => {
2476         const opt = document.createElement("option");
2477         opt.value = l.id;
2478         opt.textContent = `${l.title}（${l.scene_name}）`;
2479         sel.appendChild(opt);
2480     });
2481     currentLesson = data.items[0] || null;
2482     if (currentLesson) sel.value = currentLesson.id;
2483     sel.onchange = async () => {
2484         currentLesson = data.items.find((x) => x.id === sel.value) || null;
2485         showBanner(`已选择课时：${currentLesson ? currentLesson.title : "-"}`);
2486         await refreshClips();
2487     };
2488     await refreshClips();
2489 }
2490 [padding]
2491 async function refreshClips() {
2492     const box = el("clips");
2493     box.innerHTML = "";
2494     currentClip = null;
2495     if (!currentLesson) return;
2496     const data = await api(`/lessons/${currentLesson.id}/clips`);
2497     data.items.forEach((c) => {
2498         const d = document.createElement("div");
2499         d.className = "clip";
2500         d.innerHTML = `<div class="t">${c.label}</div><div class="m">${c.end_ms

```

```

2501 - c.start_ms) / 1000}s · 默认倍速 ${c.default_speed}x</div>`;
2502     d.onclick = () => {
2503         currentClip = c;
2504         qsa(".clip").forEach((x) => x.classList.remove("active"));
2505         d.classList.add("active");
2506         showBanner(`已选择片段: ${c.label}`);
2507     };
2508     box.appendChild(d);
2509 });
2510 }
2511 [padding]
2512 function renderPolicy(policy) {
2513     if (!policy) {
2514         el("policy").textContent = "";
2515         return;
2516     }
2517     const p = policy.capabilities;
2518     el("policy").textContent = `场景策略: 强制同步=${p.force_sync ? "是" :
2519 "否"}; 学生跳转=${p.student_seek ? "允许" : "禁止"}; 倍速=${p.student_speed
2520 ? "允许" : "禁止"}; 循环=${p.student_loop ? "允许" : "禁止"}`;
2521 }
2522 [padding]
2523 function pushEvent(evt) {
2524     const tl = el("timeline");
2525     const row = document.createElement("div");
2526     row.className = "evt";
2527     row.innerHTML = `<div class="dot"></div><div class="c"><div class=
2528 "ts">${fmtTs(evt.created_at)}</div><div class="msg">${evt.action} ·
2529 ${evt.summary}</div></div>`;
2530     tl.prepend(row);
2531 }
2532 [padding]
2533 async function pollEvents() {
2534     if (!currentSession) return;
2535     const data = await api(`/sessions/${currentSession.id}/events?since=
2536 ${since}`);
2537     data.items.forEach((e) => {
2538         since = Math.max(since, e.seq);
2539         pushEvent(e);
2540     });
2541     setTimeout(pollEvents, 900);
2542 }
2543 [padding]
2544 async function ensureSessionPolicy() {
2545     if (!currentSession) return;
2546     const data = await api(`/sessions/${currentSession.id}/policy`);
2547     renderPolicy(data.policy);
2548 }
2549 [padding]
2550 function requireTeacher() {

```

```

2551     const { user } = getAuth();
2552     if (!user || user.role !== "teacher") {
2553         showTipKind("warning", "请先用 teacher 登录。");
2554         return false;
2555     }
2556     return true;
2557 }
2558 [padding]
2559 function requireClip() {
2560     if (!requireTeacher()) return false;
2561     if (!currentSession) {
2562         showTipKind("warning", "请先开始上课。");
2563         return false;
2564     }
2565     if (!currentClip) {
2566         showTipKind("warning", "请选择片段。");
2567         return false;
2568     }
2569     return true;
2570 }
2571 [padding]
2572 async function main() {
2573     installShotTag();
2574     clearBanner();
2575     const { user } = getAuth();
2576     if (user) el("who").textContent = `已登录: ${user.name} (${user.role})`;
2577 [padding]
2578     await refreshScenes();
2579     await refreshLessons();
2580 [padding]
2581     el("btnLogin").onclick = async () => {
2582         const u = el("u").value.trim();
2583         const p = el("p").value.trim();
2584         const data = await api("/login", { method: "POST", body:
2585 JSON.stringify({ username: u, password: p }) });
2586         setAuth(data.token, data.user);
2587         el("who").textContent = `已登录: ${data.user.name} (${data.user.role})`;
2588 [padding]
2589         showBanner(`登录成功: ${data.user.role}`);
2590     };
2591 [padding]
2592     el("btnAddLesson").onclick = async () => {
2593         if (!requireTeacher()) return;
2594         const title = `Unit ${Math.floor(Math.random() * 6) + 1} 听力训练`;
2595         const sceneId = el("sceneSelect").value;
2596         const data = await api("/lessons", { method: "POST", body:
2597 JSON.stringify({ title, scene_id: sceneId, teacher_id: "u_teacher",
2598 book_unit: "Demo" }) });
2599         currentLesson = data.item;
2600         await refreshLessons();

```

```
2601     showBanner(`已新增课时: ${currentLesson.title}`);
2602 };
2603 [padding]
2604     el("btnEditLesson").onclick = async () => showTipKind("warning",
2605 "演示版: 编辑课时入口 (可扩展为表单)。");
2606     el("btnDeleteLesson").onclick = async () => showTipKind("danger",
2607 "演示版: 删除课时确认 (示例)。");
2608 [padding]
2609     el("btnStart").onclick = async () => {
2610         if (!requireTeacher()) return;
2611         if (!currentLesson) return;
2612         const data = await api(`/lessons/${currentLesson.id}/start`, { method:
2613 "POST", body: JSON.stringify({}) });
2614         currentSession = data.session;
2615         since = 0;
2616         el("timeline").innerHTML = "";
2617         await ensureSessionPolicy();
2618         pollEvents();
2619         showBanner(`课堂已开始: session=${currentSession.id}`);
2620     };
2621 [padding]
2622     el("btnEnd").onclick = async () => {
2623         if (!requireTeacher()) return;
2624         if (!currentSession) return;
2625         await api(`/sessions/${currentSession.id}/end`, { method: "POST", body:
2626 JSON.stringify({}) });
2627         showBanner("已结束上课");
2628         currentSession = null;
2629     };
2630 [padding]
2631     el("btnPlay").onclick = async () => {
2632         if (!requireClip()) return;
2633         await api(`/sessions/${currentSession.id}/event`, { method: "POST",
2634 body: JSON.stringify({ action: "play", actor_user_id: "u_teacher", target:
2635 "all_students", payload: { clip_id: currentClip.id, label: currentClip.label
2636 } }) });
2637     };
2638     el("btnPause").onclick = async () => {
2639         if (!requireTeacher() || !currentSession) return;
2640         await api(`/sessions/${currentSession.id}/event`, { method: "POST",
2641 body: JSON.stringify({ action: "pause", actor_user_id: "u_teacher", target:
2642 "all_students", payload: { } }) });
2643     };
2644     el("btnLoop").onclick = async () => {
2645         if (!requireClip()) return;
2646         await api(`/sessions/${currentSession.id}/event`, { method: "POST",
2647 body: JSON.stringify({ action: "ab_loop", actor_user_id: "u_teacher",
2648 target: "all_students", payload: { clip_id: currentClip.id, a_ms:
2649 currentClip.start_ms, b_ms: currentClip.end_ms } }) });
2650     };
```

```
2651     el("btnSpeed").onclick = async () => {
2652         if (!requireClip()) return;
2653         await api(`/sessions/${currentSession.id}/event`, { method: "POST",
2654 body: JSON.stringify({ action: "speed", actor_user_id: "u_teacher", target:
2655 "all_students", payload: { clip_id: currentClip.id, speed: 1.2 } }) });
2656     };
2657 [padding]
2658     el("btnAddClip").onclick = async () => showTipKind("warning",
2659 "演示版：新增片段入口（可扩展为表单）。");
2660     el("btnEditClip").onclick = async () => showTipKind("warning",
2661 "演示版：编辑片段入口（可扩展为表单）。");
2662     el("btnDeleteClip").onclick = async () => showTipKind("warning",
2663 "演示版：删除片段确认（示例）。");
2664 }
2665 [padding]
2666 main().catch((err) => {
2667     console.error(err);
2668     showTipKind("warning", "页面初始化失败： " + err.message);
2669 });
2670 [padding]
2671 # scripts/capture_manual_screenshots.mjs
2672 import fs from "node:fs";
2673 import path from "node:path";
2674 import { pathToFileURL } from "node:url";
2675 import { spawnSync } from "node:child_process";
2676 [padding]
2677 const root = process.cwd();
2678 const outDir = path.join(root, "images");
2679 const baseUrl = String(process.env.SCREENSHOT_BASE_URL || "").trim();
2680 const edgePath = String(process.env.EDGE_EXECUTABLE_PATH || "").trim();
2681 [padding]
2682 if (!baseUrl) {
2683     throw new Error("Missing SCREENSHOT_BASE_URL env var.");
2684 }
2685 if (!edgePath) {
2686     throw new Error("Missing EDGE_EXECUTABLE_PATH env var.");
2687 }
2688 [padding]
2689 function resolvePlaywrightEntry() {
2690     const candidate = path.resolve(root, "..", "node_modules", "playwright",
2691 "index.mjs");
2692     if (fs.existsSync(candidate)) return candidate;
2693     const alt = path.resolve(root, "..", "..", "node_modules", "playwright",
2694 "index.mjs");
2695     if (fs.existsSync(alt)) return alt;
2696     throw new Error("Cannot locate playwright/index.mjs under workspace
2697 node_modules.");
2698 }
2699 [padding]
2700 const playwrightEntry = resolvePlaywrightEntry();
```

```
2701 const { chromium } = await import(pathToFileURL(playwrightEntry).href);
2702 [padding]
2703 fs.mkdirSync(outDir, { recursive: true });
2704 [padding]
2705 function freshPath(filename) {
2706     const p = path.join(outDir, filename);
2707     try {
2708         fs.rmSync(p, { force: true });
2709     } catch {
2710         // ignore
2711     }
2712     return p;
2713 }
2714 [padding]
2715 function alignCreationTimeToWriteTime() {
2716     // Explorer "日期" column may be configured as "创建日期". When a file is
2717     overwritten in-place,
2718     // CreationTime can stay old. Aligning avoids "not latest" confusion.
2719     const cmd = [
2720         "-NoProfile",
2721         "-NonInteractive",
2722         "-Command",
2723         [
2724             `$dir='${outDir.replace('/', 'g', '\\')}';`,
2725             "Get-ChildItem -LiteralPath $dir -Filter *.png | ForEach-Object {",
2726             "$_.CreationTime = $_.LastWriteTime;",
2727             "}",
2728             "Write-Output 'creation_time_aligned'",
2729         ].join(" "),
2730     ];
2731     spawnSync("powershell", cmd, { stdio: "ignore" });
2732 }
2733 [padding]
2734 async function loginAs(page, username) {
2735     await page.goto(baseUrl, { waitUntil: "domcontentloaded" });
2736     await page.goto(`${baseUrl}teacher.html`, { waitUntil: "domcontentloaded" });
2737 };
2738 await page.waitForSelector("#sceneSelect", { timeout: 10000 });
2739 await page.fill("#u", username);
2740 await page.fill("#p", "demo");
2741 await page.click("#btnLogin");
2742 await page.waitForTimeout(600);
2743 }
2744 [padding]
2745 async function selectScene(page, index) {
2746     const sel = page.locator("#sceneSelect");
2747     await sel.selectOption({ index });
2748     await page.waitForTimeout(300);
2749     await page.waitForSelector(".device", { timeout: 10000 });
2750 }
```

```

2751 [padding]
2752 async function selectFirstLesson(page) {
2753   await page.waitForTimeout(300);
2754   const lesson = page.locator("#lessonSelect");
2755   await page.waitForFunction(() => {
2756     const el = document.getElementById("lessonSelect");
2757     return !!el && el.querySelectorAll("option").length > 0;
2758   });
2759   const count = await lesson.locator("option").count();
2760   if (count > 0) {
2761     await lesson.selectOption({ index: 0 });
2762   }
2763   await page.waitForTimeout(300);
2764   await page.waitForSelector(".clip", { timeout: 10000 });
2765 }
2766 [padding]
2767 async function selectFirstClip(page) {
2768   await page.waitForSelector(".clip", { timeout: 10000 });
2769   const clip = page.locator(".clip").first();
2770   await clip.click();
2771   await page.waitForTimeout(300);
2772 }
2773 [padding]
2774 async function main() {
2775   const browser = await chromium.launch({ headless: true, executablePath:
2776     edgePath });
2777   const page = await browser.newPage({ viewport: { width: 1600, height: 1000
2778     } });
2779   [padding]
2780   // 01: login form (before login) — visually distinct from post-login.
2781   await page.goto(`${baseUrl}teacher.html`, { waitUntil: "domcontentloaded" }
2782 );
2783   await page.waitForSelector("#btnLogin", { timeout: 10000 });
2784   await page.waitForFunction(() => typeof window.__shotTag === "function");
2785   await page.evaluate(() => window.__shotTag("截图 01: 登录页（未登录状态）"));
2786 ;
2787   await page.screenshot({ path: freshPath("01_login.png"), fullPage: true });
2788 [padding]
2789 [padding]
2790   // 02: after login (teacher)
2791   await loginAs(page, "teacher");
2792   await page.waitForFunction(() => {
2793     const who = document.getElementById("who");
2794     return who && who.textContent && who.textContent.includes("teacher");
2795   });
2796   await page.waitForFunction(() => typeof window.__shotTag === "function");
2797   await page.evaluate(() => window.__shotTag("截图 02: 登录成功与场景选择"));
2798   await page.screenshot({ path: freshPath("02_scene_select.png"), fullPage:
2799     true });
2800 [padding]

```



```
2801     await selectScene(page, 1);
2802     await selectScene(page, 2);
2803     await page.waitForFunction(() => typeof window.__shotTag === "function");
2804     await page.evaluate(() => window.__shotTag("截图 03： 设备面板（不同场景）"))
2805     ;
2806     await page.locator("#devices").screenshot({ path:
2807     freshPath("03_devices.png") });
2808     [padding]
2809     await selectFirstLesson(page);
2810     await page.click("#btnAddLesson");
2811     await page.waitForTimeout(600);
2812     await page.waitForFunction(() => typeof window.__shotTag === "function");
2813     await page.evaluate(() => window.__shotTag("截图 04： 新增课时后的列表"));
2814     await page.locator(".card:has(#lessonSelect)").screenshot({ path:
2815     freshPath("04_lessons.png") });
2816     [padding]
2817     // Ensure clip region is visible and capture after selecting a clip
2818     await selectFirstLesson(page);
2819     await selectFirstClip(page);
2820     await page.waitForFunction(() => typeof window.__shotTag === "function");
2821     await page.evaluate(() => window.__shotTag("截图 05： 片段列表与选中高亮"));
2822     await page.screenshot({ path: freshPath("05_clips.png"), fullPage: true });
2823     [padding]
2824     [padding]
2825     await page.click("#btnStart");
2826     await page.waitForTimeout(600);
2827     await page.waitForSelector("#policy", { timeout: 10000 });
2828     await page.waitForFunction(() => typeof window.__shotTag === "function");
2829     await page.evaluate(() => window.__shotTag("截图 06： 开始上课（会话策略）"))
2830     ;
2831     await page.locator("#policy").screenshot({ path: freshPath("06_start.png")
2832     });
2833     [padding]
2834     await selectFirstClip(page);
2835     await page.waitForFunction(() => typeof window.__shotTag === "function");
2836     await page.evaluate(() => window.__shotTag("截图 07： 课堂控制台按钮区（已选
2837     片段）"));
2838     await page.locator(".controls").first().screenshot({ path:
2839     freshPath("07_controls.png") });
2840     [padding]
2841     await page.click("#btnPlay");
2842     await page.waitForTimeout(300);
2843     await page.click("#btnLoop");
2844     await page.waitForTimeout(300);
2845     await page.click("#btnSpeed");
2846     await page.waitForTimeout(300);
2847     await page.waitForFunction(() => {
2848     const tl = document.getElementById("timeline");
2849     return tl && tl.querySelectorAll(".evt").length >= 2;
2850     });
```

```
2851     await page.waitForFunction(() => typeof window.__shotTag === "function");
2852     await page.evaluate(() => window.__shotTag("截图 08： 时间轴日志（已有事件）
2853     "));
2854     await page.screenshot({ path: freshPath("08_timeline.png"), fullPage: true
2855     });
2856     [padding]
2857     await page.click("#btnDeleteLesson");
2858     await page.waitForSelector("#modal:not(.hidden)", { timeout: 3000 });
2859     await page.waitForFunction(() => typeof window.__shotTag === "function");
2860     await page.evaluate(() => window.__shotTag("截图 09： 删除课时确认提示"));
2861     await page.screenshot({ path: freshPath("09_confirm_delete_lesson.png"),
2862     fullPage: true });
2863     await page.evaluate(() => window.__rkModal?.closeModal?.());
2864     await page.waitForTimeout(250);
2865     [padding]
2866     await page.click("#btnDeleteClip");
2867     await page.waitForSelector("#modal:not(.hidden)", { timeout: 3000 });
2868     await page.waitForFunction(() => typeof window.__shotTag === "function");
2869     await page.evaluate(() => window.__shotTag("截图 10： 删除片段确认提示"));
2870     await page.screenshot({ path: freshPath("10_confirm_delete_clip.png"),
2871     fullPage: true });
2872     [padding]
2873     // 11: admin policy editor
2874     await page.goto(`${baseUrl}admin.html`, { waitUntil: "domcontentloaded" });
2875     [padding]
2876     await page.waitForSelector("#sceneSelect", { timeout: 10000 });
2877     await page.waitForFunction(() => typeof window.__shotTag === "function");
2878     await page.evaluate(() => window.__shotTag("截图 11： 管理端策略配置"));
2879     await page.screenshot({ path: freshPath("11_admin_policy.png"), fullPage:
2880     true });
2881     [padding]
2882     // 12: student follow view (use currentSession if available)
2883     await page.goto(`${baseUrl}student.html`, { waitUntil: "domcontentloaded" }
2884     );
2885     await page.waitForSelector("#sessionId", { timeout: 10000 });
2886     // If no session id, still capture the student page layout.
2887     await page.waitForFunction(() => typeof window.__shotTag === "function");
2888     await page.evaluate(() => window.__shotTag("截图 12： 学生端跟随入口"));
2889     await page.screenshot({ path: freshPath("12_student_follow.png"),
2890     fullPage: true });
2891     [padding]
2892     await browser.close();
2893     alignCreationTimeToWriteTime();
2894     }
2895     [padding]
2896     main().catch((err) => {
2897     console.error(err);
2898     process.exit(1);
2899     });
2900     # scripts/count_lines.py
```

```
2901 """Count non-empty source-code lines for the generated project.
2902 [padding]
2903 This is a project-local copy used by the gated pipeline.
2904 """
2905 [padding]
2906 from __future__ import annotations
2907 [padding]
2908 from pathlib import Path
2909 [padding]
2910 [padding]
2911 def _resolve_repo_root() -> Path:
2912     cwd = Path.cwd().resolve()
2913     if (cwd / "apps").exists() or (cwd / "packages").exists():
2914         return cwd
2915     return Path(__file__).resolve().parents[1]
2916 [padding]
2917 [padding]
2918 REPO_ROOT = _resolve_repo_root()
2919 SOURCE_ROOTS = [REPO_ROOT / "apps", REPO_ROOT / "packages", REPO_ROOT /
2920 "scripts"]
2921 TARGET_SUFFIXES = {".py", ".js", ".mjs", ".css", ".html", ".sql", ".json"}
2922 EXCLUDED_DIRS = {
2923     ".git",
2924     ".tools",
2925     "__pycache__",
2926     "node_modules",
2927     "run_reports",
2928     "tmp",
2929     "copyright-application-materials",
2930 }
2931 [padding]
2932 [padding]
2933 def iter_source_files() -> list[Path]:
2934     files: list[Path] = []
2935     for root in SOURCE_ROOTS:
2936         if not root.exists():
2937             continue
2938         for path in root.rglob("*"):
2939             if path.is_dir():
2940                 continue
2941             if any(part in EXCLUDED_DIRS for part in path.parts):
2942                 continue
2943             if path.suffix.lower() not in TARGET_SUFFIXES:
2944                 continue
2945             files.append(path)
2946     return sorted(files)
2947 [padding]
2948 [padding]
2949 def count_lines() -> int:
2950     total = 0
```

```

2951     for path in iter_source_files():
2952         try:
2953             content = path.read_text(encoding="utf-8")
2954         except UnicodeDecodeError:
2955             continue
2956         total += sum(1 for line in content.splitlines() if line.strip())
2957     return total
2958 [padding]
2959 [padding]
2960 if __name__ == "__main__":
2961     print(count_lines())
2962 [padding]
2963 # scripts/generate_demo_library.py
2964 # -*- coding: utf-8 -*-
2965 from __future__ import annotations
2966 [padding]
2967 import json
2968 from pathlib import Path
2969 [padding]
2970 [padding]
2971 def main() -> int:
2972     out = Path("apps/api/data/audio_library.json")
2973     out.parent.mkdir(parents=True, exist_ok=True)
2974     items = []
2975     # 生成较大的演示音频库：用于展示“资源集中管理”，同时满足行数门槛。
2976     # 每条记录一行写入（pretty line-by-line），便于在源代码文档里分页。
2977     for i in range(1, 2601):
2978         unit = (i % 24) + 1
2979         scene_tag = ["classroom", "language_lab", "live", "corner",
2980 "selfstudy"][i % 5]
2981         items.append(
2982             {
2983                 "id": f"asset_{i:04d}",
2984                 "title": f"Unit {unit} - Listening Clip {i:04d}",
2985                 "duration_sec": 18 + (i % 55),
2986                 "level": ["A1", "A2", "B1", "B2"][i % 4],
2987                 "scene_tag": scene_tag,
2988                 "source_uri": f"/static/audio/unit{unit:02d}/clip_{i:04d}
2989 .mp3",
2990             }
2991         )
2992     out.write_text("\n".join(json.dumps(x, ensure_ascii=False) for x in
2993 items), encoding="utf-8")
2994     print(str(out.resolve()))
2995     return 0
2996 [padding]
2997 [padding]
2998 if __name__ == "__main__":
2999     raise SystemExit(main())
3000 [padding]

```